SPE-230042-MS

Fast-Track Deployment of a Modular Water Handling Unit to Restore Production in a High Water-Cut Field: A Case Study from Bisat, Oman

Y. Z. Al Nabhani, Business Administration, Heriot-Watt University, Edinburgh, UK / Chemical Engineer, Sheffield University, Sheffield, UK / OQEP, Muscat, Sultanate of Oman

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

As many oil fields in Oman mature, a significant operational challenge has emerged: the rapid increase in water cut, often exceeding 70% and, in some cases, reaching over 90%. This sharp rise in produced water volumes places immense pressure on existing surface facilities, many of which were not originally designed to manage such a high water influx. Consequently, traditional oil separation and processing infrastructure has become a bottleneck, constraining production and leading to the shut-in of otherwise productive wells.

In this context, water handling has shifted from being a secondary concern to becoming the dominant operational focus in the effort to sustain oil production. The situation is further exacerbated by tightening environmental regulations issued by Oman's Ministry of Environment and Climate Affairs (MECA), which mandate improved treatment, reinjection, or reuse of produced water. Operators are increasingly required to deliver operational excellence while aligning with Oman Vision 2040, which emphasizes resource sustainability, energy efficiency, and environmental stewardship.

This paper presents a practical, field-proven case study of the Bisat Field, where a fast-tracked Water Handling Unit (WHU) was deployed to alleviate severe production constraints caused by high water cut. Within just three months, the WHU solution enabled the reactivation of over 150 shut-in wells, restored approximately 5,000 barrels of oil per day (BOPD), and improved the utilization of existing facilities by processing the additional oil recovered through the WHU. The rapid execution, modular design, and integration of mobile systems allowed the operation to bypass traditional infrastructure expansion timelines, offering a replicable model for similar challenges.

Introduction

As many oil fields in Oman mature, a significant operational challenge has emerged: the rapid increase in water cut, often exceeding 70% and, in some cases, reaching over 90%. This sharp rise in produced water volumes places immense pressure on existing surface facilities, many of which were not originally designed to manage such a high water influx. Consequently, traditional oil separation and processing infrastructure has become a bottleneck, constraining production and leading to the shut-in of otherwise productive wells.

In this context, water handling has shifted from being a secondary concern to becoming the dominant operational focus in the effort to sustain oil production. The situation is further exacerbated by tightening environmental regulations issued by Oman's Ministry of Environment and Climate Affairs (MECA), which mandate improved treatment, reinjection, or reuse of produced water. Operators are increasingly required to deliver operational excellence while aligning with Oman Vision 2040, which emphasizes resource sustainability, energy efficiency, and environmental stewardship.

This paper presents a practical, field-proven case study of the Bisat Field, where a fast-tracked Water Handling Unit (WHU) was deployed to alleviate severe production constraints caused by high water cut. Within just three months, the WHU solution enabled the reactivation of over 150 shut-in wells, restored approximately 5,000 barrels of oil per day (BOPD), and improved the utilization of existing facilities by processing the additional oil recovered through the WHU. The rapid execution, modular design, and integration of mobile systems allowed the operation to bypass traditional infrastructure expansion timelines, offering a replicable model for similar challenges.

The objective of this paper is to demonstrate how a fit-for-purpose, modular WHU solution can be executed under fast-track conditions to support mature field operations. It highlights key technical, operational, and strategic enablers, from early vendor engagement to digital integration, that made this deployment successful. The insights gained offer practical value to operators facing comparable water handling constraints in high water-cut fields across the region and beyond.

Field Background & Problem Statement

The Bisat Field, located in Block 60 of Oman, is a multi-phase development comprising three key stages: Bisat A, Bisat B, and Bisat C. These phases were brought online with progressively higher water handling capacities, 60,000 barrels per day (BPD) for Bisat A, 125,000 BPD for Bisat B, and 330,000 BPD for Bisat C, culminating in a total facility capacity exceeding 400,000 BPD of fluids. Designed oil production from these stages was expected to reach approximately 12,000, 25,000, and 30,000 BOPD respectively.

However, as the field matured, the water cut rose sharply, reaching levels of approximately 91%. This extreme increase in produced water volumes significantly overwhelmed the surface processing capacity. The existing oil separation infrastructure, although designed for large volumes, was not optimized for handling such a high water-to-oil ratio. As a result, the plant's oil design capacity became underutilized, and over 150 high-water-cut wells were shut in, directly impacting production targets and economic performance.

The escalation in water cut significantly reduced oil throughput and introduced a range of operational, environmental, and financial challenges. The existing surface facilities were not equipped to manage the high water-to-oil ratio, resulting in capacity bottlenecks and increasing the risk of production deferment and unplanned downtime. At the same time, the operation had to maintain environmental compliance with regulations set by Oman's Ministry of Environment and Climate Affairs (MECA), particularly regarding water treatment and reinjection. These compounding pressures highlighted the urgent need for a rapid surface facility intervention to sustain production while permanent facility upgrades were still in progress.

Against this backdrop, it became clear that relying on conventional facility upgrades alone, often requiring 10 to 18 months from design to commissioning, would be insufficient to meet near-term production needs. An interim strategy was necessary to bridge the operational gap and restore output while longer-term expansions, such as the Bisat C upgrade, were still underway.

Challenges in Handling High Water-Cut Fields

In mature oil fields like Bisat, where water cuts can exceed 90%, traditional surface facilities often struggle to manage the high water-to-oil ratio. These systems were originally designed to accommodate a forecasted water cut range based on expected production profiles. However, as actual water cut levels escalated far beyond design assumptions, reaching up to 91% in some wells, the surface facilities faced significant strain in terms of separation, treatment, and reinjection capacity.

While Bisat was already progressing toward a major facility expansion, the scale of the water cut escalation outpaced the readiness of the permanent infrastructure. Constructing or upgrading conventional water handling systems typically requires 10 to 18 months due to long engineering, procurement, and construction lead times, a timeline that does not support immediate production recovery.

In addition to the high overall water cut, a major constraint was the uneven distribution of water handling capacity across the Bisat field's phased infrastructure. The smaller stations, Bisat A and Bisat B, reached their maximum water handling capacity well before achieving their oil production potential. In some cases, these stations operated at under 60% of their oil design capacity simply because the water component in the fluid stream exceeded what the facilities could process. As a result, high-water-cut wells had to be shut in, leaving valuable oil volumes untapped.

This imbalance also introduced significant operational instability. With the water systems operating near their design limits, even minor feed fluctuations, such as commissioning of new wells, well restarts after trips, or scheduled maintenance, could overwhelm the system and trigger facility shutdowns. These frequent trips not only disrupted production continuity but also increased strain on operating teams and reduced overall facility efficiency.

Technical complexity also played a role. High water cut fluids in Bisat exhibited variable properties, including emulsions, solids, and salinity, which reduce the effectiveness of traditional separation systems. This complexity, combined with the tight environmental standards set by Oman's Ministry of Environment and Climate Affairs (MECA), made fast and reliable water separation and disposal essential.

Given these challenges, Bisat required a rapid, modular, and operationally flexible surface solution, one that could be deployed quickly, integrated safely into the existing production setup, and scaled as needed until the permanent facility expansion was completed. The Water Handling Unit (WHU) approach provided an ideal fit for this context.

WHU Solution Design and Deployment Rationale

To address the operational bottlenecks caused by extreme water cut in the Bisat Field, a fit-for-purpose Water Handling Unit (WHU) was designed and deployed as a temporary surface solution. The goal was to restore production from shut-in wells, relieve pressure on overburdened facilities, and maintain oil quality and export specifications, all within a fast-track execution window.

The WHU was engineered to handle 50,000 barrels of fluid per day, comprising 45,000 BWPD of produced water and 5,000 BOPD of crude oil, along with up to 1 MMSCFD of associated gas. The unit was designed for full compliance with operational and environmental requirements, with outlet specifications of:

- <2% water cut in exported oil,
- <100 ppm oil-in-water (OIW) in the separated water stream, and
- Reinjection pressure of 82–85 bar for water disposal,
- Export pressure of 21 bar for treated oil.

As it can be seen from Figure 1 above that the facility configuration included two processing trains (Train A and Train B), each equipped with a first-stage three-phase separator (78" ID × 34 ft) for bulk separation of oil, water, and gas. The separated oil passed through an indirect water bath heater for viscosity management before entering a second-stage two-phase separator for degassing. The final treated oil, containing less than 2% water, was routed to six storage tanks, then transferred via crude transfer pumps at 25 barg to the existing Bisat B station for further treatment and final export to the main oil line. This approach also leveraged existing infrastructure for both oil and water handling, by sending oil to Bisat B and routing produced water to the field's established disposal network, which significantly expedited deployment by avoiding the need for new sinks.

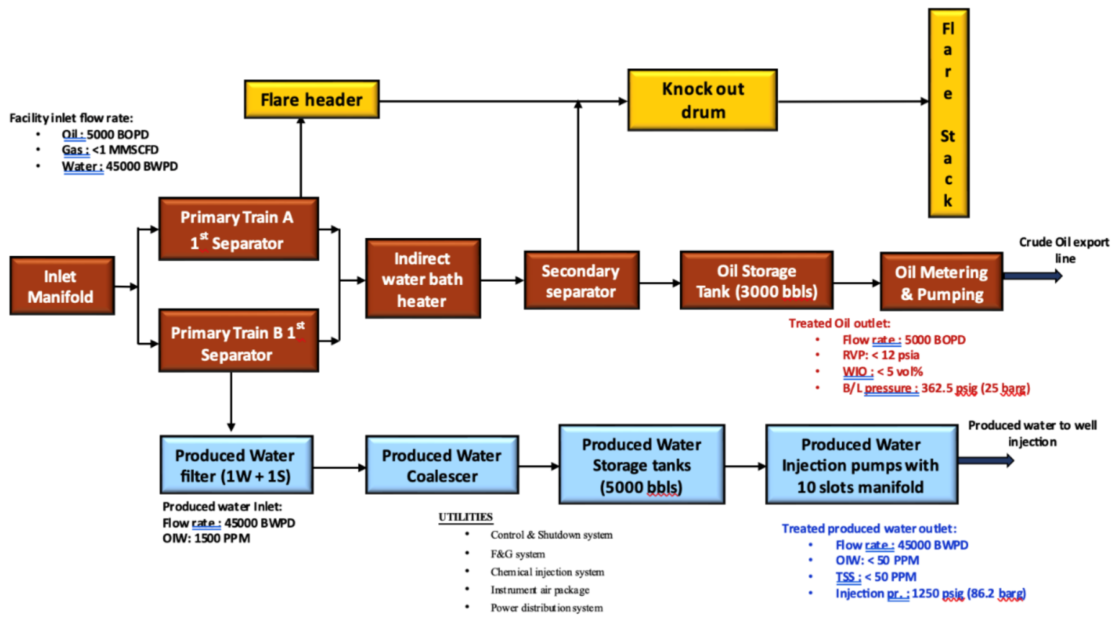

Figure 1—WHU Process Flow Diagram

The produced water from both trains underwent filtration and coalescing treatment to reduce solids and residual oil content before being stored in ten dedicated tanks and reinjected into designated wells using existing high-pressure injection pumps. All systems were compliant with NACE MR0175 standards, ensuring material compatibility even though no H_2S was present in the fluid composition. The gas stream was routed to the HP flare system, ensuring safe handling of associated gas.

Importantly, the WHU was designed for modularity, mobility, and fast deployment. Key components, including separators, coalescers, filtration units, pumps, tanks, and control systems, were delivered as skid-mounted plug-and-play modules, minimizing civil work and tie-in complexity. This allowed for the entire facility to be mobilized and operational within three months, bridging the production gap while the Bisat C permanent expansion project progressed toward completion.

The WHU was fully integrated with the existing production infrastructure and control systems, enabling coordinated operation with the upstream wells and the central oil processing facility. This integration allowed the WHU to dynamically adjust to changes in flow rates and pressure, improving system resilience during well trips or planned interventions.

Deployment Strategy and Execution

The success of the WHU deployment at Bisat was rooted in a fast-track execution strategy, driven by clear production imperatives and strong cross-functional collaboration. Recognizing the urgency imposed by rising water cut and constrained facility capacity, the project team adopted a streamlined execution model to minimize lead times and operational disruption.

From the outset, the deployment strategy prioritized early vendor engagement, enabling parallel progress on engineering review, procurement, and site readiness. The project leveraged ready-made modular units, including separators, filtration units, storage tanks, and pumps, which were pre-designed and tested by the vendor. This approach eliminated long fabrication lead times and allowed focus to shift toward on-site preparation and piping integration.

The only major fabrication activities involved the interconnection piping between units, which was executed efficiently on-site. To accelerate deployment and reduce construction effort, LLRTP (Lightweight, Leak-Resistant Thermoplastic Pipe) was selected for the feed, export, and injection lines. This choice offered benefits in terms of corrosion resistance, ease of installation, and pressure containment, particularly for water injection at up to 85 bar.

A critical enabler was the use of skid-mounted, plug-and-play equipment, which significantly reduced civil work requirements and allowed for faster installation. Modules were delivered to site with standardized hook-up points, allowing the mechanical, electrical, and instrumentation teams to complete integration with minimal interference to ongoing operations.

Site readiness was carefully managed. Tie-in points to upstream manifolds and downstream transfer infrastructure (e.g., Bisat B station) were selected for accessibility and operational safety. Utilities, including power, instrument air, chemical injection, and control interfaces, were prepared in parallel to equipment delivery to minimize delays.

The entire process, from contract call-out to commissioning, was completed within approximately three months, far faster than the typical 10-18 month cycle for conventional brownfield water handling projects. The timeline included:

- Tender release and contract award
- Vendor mobilization and delivery of WHU modules
- On-site piping fabrication and unit interconnections
- Commissioning and startup of Train A and Train B
- Progressive increase in handling from 15,000 to 45,000 BWPD

The ability to utilize existing oil export infrastructure and water injection systems further reduced installation complexity and contributed to the compressed project schedule

Throughout execution, a risk-based approach was adopted to mitigate challenges typical of fast-track projects. Potential issues, including interface mismatches, start-up surges, and injection system backpressure, were addressed through pre-commissioning reviews, vendor SATs (Site Acceptance Tests), and a phased startup strategy. Close coordination with field teams enabled dynamic adjustments to live feed fluctuations, well restarts, and operational upsets.

The WHU was fully integrated with the central control system, allowing real-time digital monitoring of pressures, flow rates, and unit status. This enabled quick intervention during feed variability or equipment trips, improving production stability and response time from the first day of operation.

Operational Results and Performance

The deployment of the Water Handling Unit (WHU) at Bisat delivered immediate and measurable improvements in field performance. Within the first few weeks of operation, the WHU enabled the reactivation of over 50 shut-in high water cut wells, significantly increasing fluid throughput and recovering approximately 5,000 barrels of oil per day (BOPD)that would have otherwise remained deferred.

The modular dual-train design ensured stable processing of up to 45,000 BWPD of produced water and 5,000 BOPD of oil. Throughout operations, the WHU consistently met all performance criteria:

- Exported oil maintained at <2% water cut,
- Treated water achieved <100 ppm oil-in-water (OIW) for reinjection
- Stable operational pressures of 82–85 bar for injection and 21–25 bar for oil export.

The recovered oil from the WHU was transferred to the Bisat B station for further treatment and export via the main oil line. While the WHU operated independently from the Bisat A/B/C processing trains, its contribution of higher-purity oil into the Bisat B system led to an improved fluid mix at the receiving facility. For example, before WHU deployment, Bisat B was approaching its water handling limit while producing minimal oil, placing it at risk of exceeding design constraints. By injecting more oil-rich fluid into Bisat B, the overall oil-to-water ratio improved, expanding the operating envelope and reducing the frequency of process upsets and trips.

Although the water volume to existing stations remained largely unchanged, the enhanced fluid composition allowed better separation efficiency, more stable operations, and improved utilization of Bisat B's design capacity. This effect also helped alleviate operational strain at the broader field level.

In parallel, the WHU reinforced environmental compliance through closed-loop water handling. All treated water was reinjected into approved disposal wells, fully aligned with MECA regulations, while separated gas was safely routed to the HP flare system. No surface discharge occurred during the unit's operation.

Overall, the WHU not only restored significant oil production but also enhanced field stability, water management, and process reliability while the permanent Bisat C expansion remained under development. Its success demonstrated the viability of modular water handling solutions for high water cut environments across the region.

Economic and Strategic Value

The WHU deployment at Bisat delivered not only technical and operational benefits but also clear economic and strategic value to the field and the wider asset management strategy. By enabling the reactivation of more than 150 shut-in high water cut wells, the project directly contributed to a recovery of approximately 5,000 BOPD, which would have otherwise remained deferred due to surface capacity constraints. This early restoration of production helped accelerate revenue generation and improved the asset's short-term cash flow performance.

Compared to conventional brownfield facility upgrades, which often require 10 to 18 months and significant engineering, civil, and utility work, the WHU was mobilized, installed, and commissioned in approximately three months. This reduced execution timeline enabled faster return on investment (ROI), as the incremental oil production began contributing to revenue within the same quarter of mobilization.

The use of ready-made modules, LLRTP piping, and existing infrastructure (e.g., Bisat B tie-in) helped minimize capital expenditure (CAPEX) while also reducing lifecycle operating costs (OPEX) through:

- Lower civil and foundation work,
- Minimal downtime during integration,
- Flexibility to relocate or scale the WHU for future applications.

Beyond financial metrics, the WHU deployment also brought strategic advantages. It provided a bridge solution that maintained production continuity while the Bisat C expansion was still under development. This ensured smoother field development phasing, avoided the risk of production decline during infrastructure transition, and protected the asset's contribution to OQ's portfolio targets.

In addition, the project supported broader in-country value (ICV) goals. Local contractors were involved in site preparation, piping fabrication, and WHU operation. The use of modular units simplified training and knowledge transfer for Omani technicians, building operational capability around fast-deploy surface solutions, an increasingly relevant skillset as many of Oman's fields mature.

The WHU concept also demonstrated scalability and adaptability, making it a viable model for replication in other high water cut environments across Block 60 and similar regional assets. Its success reinforced the importance of operational agility, modular infrastructure, and fast-track deployment models in maximizing late-life asset value and aligning with Oman's Vision 2040 goals for resource efficiency and energy sector innovation.

Discussion

The Bisat WHU case provides a valuable demonstration of how modular, fast-track surface solutions can help operators sustain production in high water cut environments where traditional infrastructure is either overloaded or still under development. While the concept of temporary or mobile water handling is not new, its successful deployment in Bisat under extreme water cut conditions, and within a two-month timeline, offers important operational insights.

One of the key takeaways is the strategic advantage of decoupling water handling from central processing systems. By isolating high WC streams and treating them through a dedicated facility, the WHU allowed the core stations to operate within a more favorable process envelope. This had a direct impact on reducing unplanned trips, improving oil-to-water ratios, and stabilizing field operations, even though the WHU did not reduce the total water volume processed by the central facilities.

Another critical success factor was the use of pre-engineered, ready-made units, which eliminated long fabrication cycles and minimized interface complexity. The modular configuration and local piping fabrication using LLRTP enabled safe and rapid tie-ins while maintaining flexibility for future relocation or reconfiguration, a clear advantage for operators managing aging assets with shifting well profiles.

The case also emphasizes the value of early vendor engagement and parallel workstreams, particularly under fast-track schedules. The combination of vendor-led equipment delivery and field-driven site preparation helped compress the execution timeline and ensured smooth commissioning without compromising operational safety or compliance.

On a broader scale, the WHU deployment at Bisat supports a growing shift in the upstream industry toward adaptive infrastructure models, where modularity, mobility, and execution agility are prioritized alongside traditional facility planning. For operators managing mature fields in Oman and across the region, such approaches offer a scalable, lower-risk path to sustain late-life production while managing water, environmental, and operational complexity.

Furthermore, this case aligns with Oman Vision 2040 goals, which emphasize innovation, efficiency, and environmental responsibility in the energy sector. By meeting MECA standards, leveraging local capabilities, and maintaining continuous production without the need for flaring or discharge, the WHU initiative reflects a practical application of these national strategic principles.

Conclusion

The fast-track deployment of a Water Handling Unit (WHU) in the Bisat Field represents a field-proven solution to one of the most pressing challenges facing mature oil fields in Oman: managing escalating water cut without compromising production continuity. Faced with surface facility limitations and over 150 shut-in wells, the operation required a rapid, modular intervention to restore oil output and maintain system stability while permanent infrastructure expansion was still underway.

The WHU enabled the recovery of approximately 5,000 barrels of oil per day, stabilized fluid processing, and ensured compliance with environmental regulations, all within a compressed execution window of approximately two months. By treating high water cut wells through a dedicated system and routing the recovered oil to existing infrastructure, the WHU not only unlocked deferred production but also improved the operating envelope of the receiving station through a better oil-water mix.

The success of the project was driven by early vendor engagement, the use of ready-made modular equipment, LLRTP-based piping, and strong coordination between engineering, operations, and commissioning teams. The local control system ensured real-time visibility and rapid response to flow variability, enhancing reliability from the outset.

Beyond immediate production gains, the Bisat WHU serves as a scalable model for similar high water cut challenges, offering both economic and strategic value. It demonstrates that with the right planning and modular design, operators can bridge critical infrastructure gaps, sustain output, and support long-term field viability, all while aligning with national goals around efficiency, environmental stewardship, and operational excellence.

References

MECA Oman. (2020). Environmental Regulations for Onshore Oil & Gas Operations. Ministry of Environment and Climate Affairs (Sultanate of Oman).